
\$and

\$or

\$nor

\$not

Logical Query Operators

DESCRIPTION

- **\$and**: Joins query clauses with a logical AND returns all documents that match the conditions of both clauses.
 - **\$not**: Inverts the effect of a query expression and returns documents that do *not* match the query expression.
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DESCRIPTION

- **\$nor**: Joins query clauses with a logical NOR returns all documents that fail to match both clauses.
- **\$or**: Joins query clauses with a logical OR returns all documents that match the conditions of either clause

Student collection

```
{
  "_id": ObjectId("6396a083f3b30b3e904d794f"),
  "name": 'Sita',
  "age": 5,
  "Hobbies": [ 'Walk', 'Cricket' ],
  "identity": { hasPanCard: false, hasAdhaarCard: true }
},
{
  "_id": ObjectId("6396a083f3b30b3e904d7950"),
  "name": 'Shyam',
  "age": 11,
  "Hobbies": [ 'Walking', 'Reading' ],
  "identity": { hasPanCard: false, hasAdhaarCard: true }
},
{
  "_id": ObjectId("6396a083f3b30b3e904d7951"),
  "name": 'Ghanshayam',
  "age": 11,
  "Hobbies": [ 'Walk', 'Cricket' ],
  "identity": { hasPanCard: false, hasAdhaarCard: true }
},
{
```

Student collection

```
{
  _id: ObjectId("6396a083f3b30b3e904d7952"),
  name: 'Rita',
  age: 11,
  Hobbies: [ 'Anime' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7953"),
  name: 'Ram',
  age: 10,
  Hobbies: [ 'Walk', 'Cricket' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7954"),
  name: 'Geeta',
  age: 12,
  Hobbies: [ 'Gaming', 'Cooking' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
```

\$or

The \$or operator performs a logical OR operation on an array of *one or more* <expressions> and selects the documents that satisfy *at least* one of the <expressions>.

Query: college>

```
db.student.find({$or:[{age:{$lte:10}},{age:$gte:12}]})
```

Output

```
{
  _id: ObjectId("6396a083f3b30b3e904d794f"),
  name: 'Sita',
  age: 5,
  Hobbies: [ 'Walk', 'Cricket' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7953"),
  name: 'Ram',
  age: 10,
  Hobbies: [ 'Walk', 'Cricket' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7954"),
  name: 'Geeta',
  age: 12,
  Hobbies: [ 'Gaming', 'Cooking' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
}
```

\$nor

\$nor performs a logical NOR operation on an array of one or more query expression and selects the documents that **fail** all the query expressions in the array

Query:

```
college> db.student.find({$nor:[{age:{$lte:10}},{age:$gte:12}]}])
```

Output

```
{
  _id: ObjectId("6396a083f3b30b3e904d7950"),
  name: 'Shyam',
  age: 11,
  Hobbies: [ 'Walking', 'Reading' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7951"),
  name: 'Ghanshayam',
  age: 11,
  Hobbies: [ 'Walk', 'Cricket' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7952"),
  name: 'Rita',
  age: 11,
  Hobbies: [ 'Anime' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
}
```

\$and

\$and performs a logical AND operation on an array of *one or more* expressions (<expression1>, <expression2>, and so on) and selects the documents that satisfy *all* the expressions.

Query:

```
college> db.student.find({$and:[{age:{$lt:11}},{hobbies:'walk'}]})
```

Output

```
{
  _id: ObjectId("6396a083f3b30b3e904d794f"),
  name: 'Sita',
  age: 5,
  Hobbies: [ 'Walk', 'Cricket' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7953"),
  name: 'Ram',
  age: 10,
  Hobbies: [ 'Walk', 'Cricket' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
}
```



\$not

\$not performs a logical NOT operation on the specified <operator-expression> and selects the documents that do *not* match the <operator-expression>. This includes documents that do not contain the field.

Query: college> db.student.find({age:{\$ne}})

Output

```
[
  {
    _id: ObjectId("6396a083f3b30b3e904d794f"),
    name: 'Sita',
    age: 5,
    Hobbies: [ 'Walk', 'Cricket' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  }
]
```

Behavior of \$and

When evaluating the clauses in the \$and expression, MongoDB's query optimizer considers which indexes are available that could help satisfy clauses of the \$and expression when selecting the best plan to execute.

To allow the query engine to optimize queries, \$and handles errors as follows:

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- If any expression supplied to `$and` would cause an error when evaluated alone, the `$and` containing the expression may cause an error but an error is not guaranteed.
 - An expression supplied after the first expression supplied to `$and` may cause an error even if the first expression evaluates to `false`
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Behavior of not:

\$not and Data Types

The operation of the \$not operator is consistent with the behavior of other operators but may yield unexpected results with some data types like arrays.

Behaviors of or:

`$or` Clauses and Indexes

When evaluating the clauses in the `$or` expression, MongoDB either performs a collection scan or, if all the clauses are supported by indexes, MongoDB performs index scans. That is, for MongoDB to use indexes to evaluate an `$or` expression, all the clauses in the `$or` expression must be supported by indexes. Otherwise, MongoDB will perform a collection scan
